



HOUSING ACCESS EQUITY

Digital Solutions for Affordable Housing



Problem

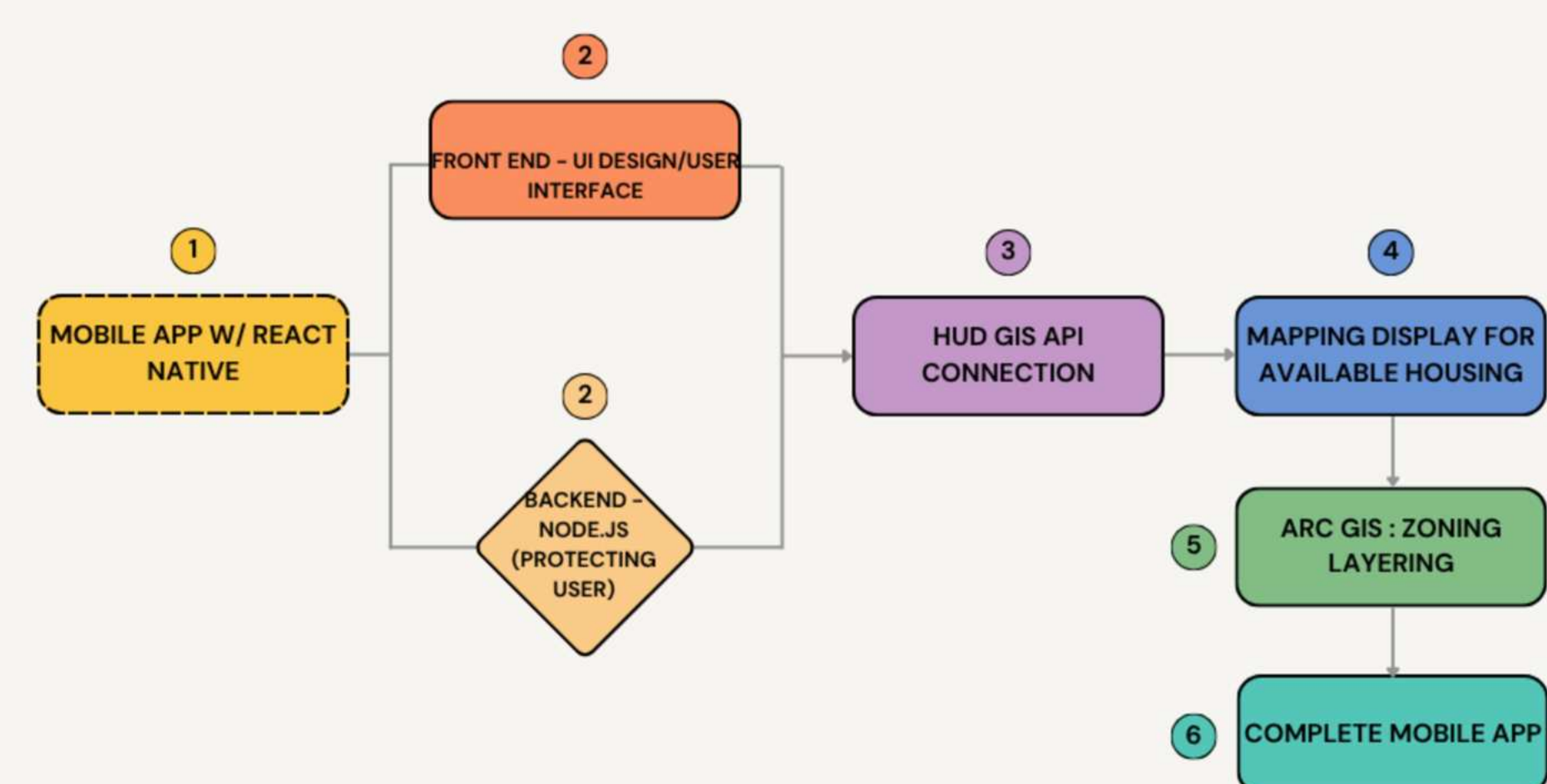
Affordable housing is one of the most pressing challenges in today's real estate market. Many people struggle to find accessible housing options, like Zillow, they often fail to cater to low-income populations or provide localized data about affordability. A centralized real estate platform that connects individuals with affordable housing opportunities, help them navigate zoning restrictions and cost limitations. This platform will help them overcome housing challenges they may face.



Purpose

This passion evolved into a heightened awareness of the structural disparities in housing availability, particularly in Sierra Leone and Ghana, where affordable housing remains constrained. In response to these observations, digital solutions for affordable housing supports my chosen solution that integrates service and technology. I aim to develop an affordable housing platform that connects individuals to accessible housing opportunities, while currently focusing in the Washington D.C area. This research addresses how a centralized platform can help individuals explore economic benefits of affordable housing.

Methodology



The project focuses on the examination of real estate, zoning regulations, HUD datasets through both frontend and backend aspects, ZIP-code data, In-App WebView, and Google Maps Street View.



Results

This centralized platform uses HUD data, ZIP-code information, and local jurisdiction resources to help match users with affordable housing opportunities. With the implementation of Google Street View and a WebView zoning showcase, it gives users a cleared understanding of neighborhoods and zoning rules.

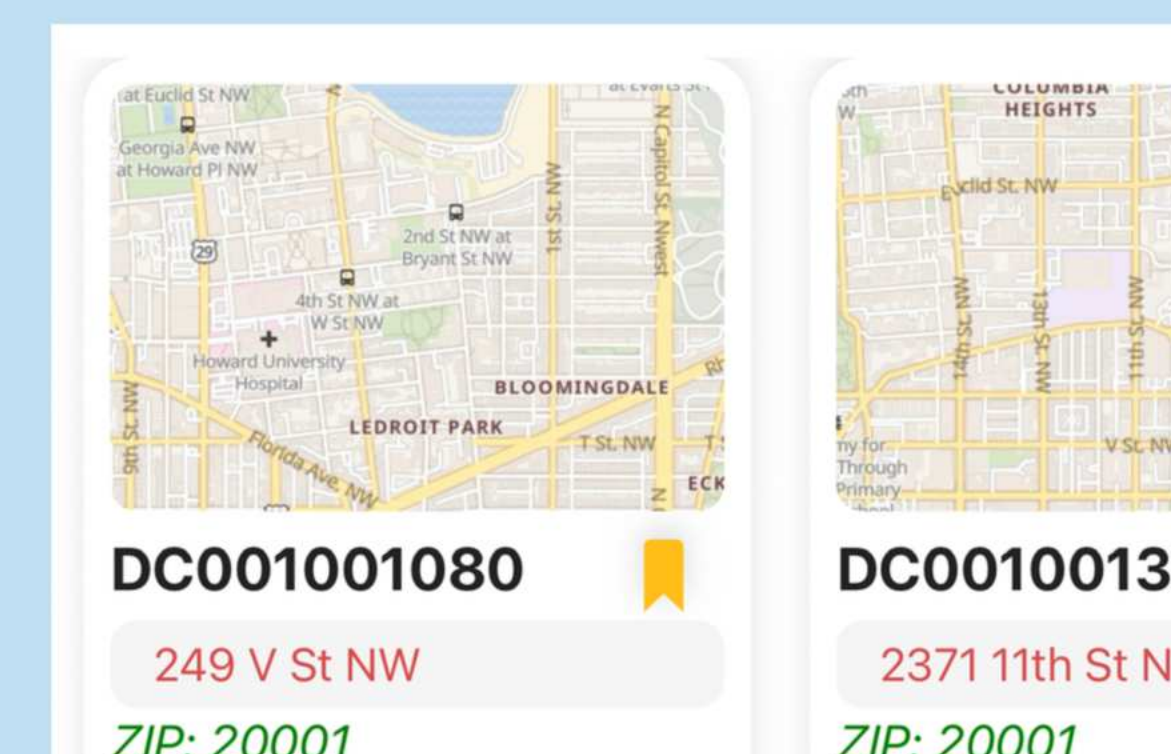
Of the many different methods, this approach was the most beneficial since React Native allows both mobile and web functionality within one hotspot (Visual Studio). This is crucial since low-income users may only be able to rely on one or the other.

Tappable addresses allow users to open location directly in google maps while also being able to interact directly in the app with Google Street View. These features help create a more centralized and practical housing search experience.

The user login screen improves access by allowing individuals to enter the platform with their own identity. HUD data helped connect affordable housing opportunities across multiple locations, making it possible to compare availability in different cities and counties.

Results showed that some areas had more affordable homes than others, with North Washington, D.C. showing a different housing pattern than Loudoun County, which is wealthier than that area of D.C. For example, one jurisdiction may offer a larger number of affordable listings, while another may have fewer units but different zoning conditions.

These results demonstrate that affordable housing access varies by location, for example Washington. D.C, Loudoun County, and Arlington County, a centralized platform can help users compare options more effectively.



Email

Password

Login

Welcome! Let's find the right house for you.

Find affordable housing near you.

20001

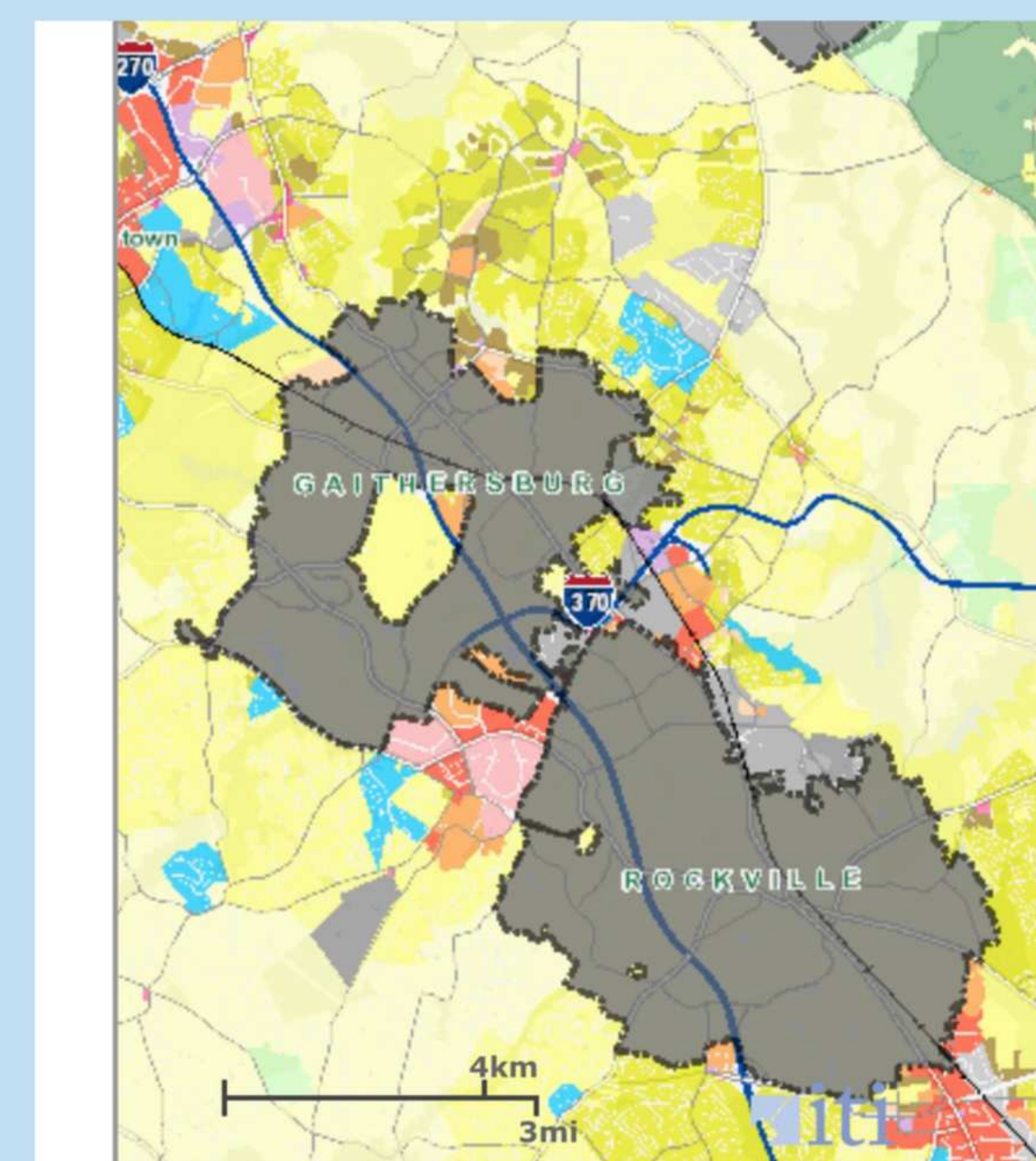
Affordable Housing Nearby



DC001001830
1227 New Jersey Ave NW
ZIP: 20001



DC001000081
915 S St NW
ZIP: 20001



Welcome! Choose a county.

Montgomery County

Zoning Explorer

Interactive Maps

Zoning Rules

Loudoun County

Fairfax County

Conclusions

This research shows that a centralized platform can improve access to affordable housing. It demonstrates how digital tools can connect users to housing data, zoning information, and location-based resources in one place. It also shows that affordable housing platforms can better support low-income families when they are designed around clarity, accessibility, and usability. In addition, expanding the platform beyond Washington, D.C, to other underserved communities strengthens its long-term impact. Overall, this work supports the idea that technology can help reduce the barriers in the housing search process and expand access to underserved communities.

Analysis

Overall, this research shows that affordable housing is affected by more than just price. Zoning restrictions, construction costs, high interest rates, and limited supply all make it harder for families to find housing they can afford. Many users also face an information problem since housing data tends to be scattered across multiple sources, making the search process confusing and inefficient. By combining HUD data, local jurisdiction resources, and a user login system, the platform creates a more organized way to searching for housing. The comparison of different locations spanning from the East Coast to West Coast also shows that affordable housing availability varies by region. This supports the need for a centralized platform that can help users make better decisions based on location, eligibility, and affordability.

References

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